

KEMAL ÖZYÖN

3rd-Year CTIS Student — Backend Systems & Applied AI

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3rd-year Computer Technology and Information Systems student (full scholarship, Bilkent University) specializing in backend systems. Builds and ships complete, working software — from a production FastAPI platform with async PostgreSQL, JWT auth, and S3 storage, to a cross-platform desktop app deployed in a real retail store. Strong mathematical foundation (calculus, linear algebra) supporting an active transition into machine learning. Seeking a backend/AI-focused internship.

EDUCATION

B.Sc. in Computer Technology and Information Systems (Full Scholarship)

Sept 2023 – Jun 2028

Bilkent University, Ankara

- Awarded a full merit-based scholarship after ranking 3,514th nationwide in the YKS university entrance exam (Quantitative Track)
- GPA: 3.24 (Last Semester: 3.55) — a consistent upward trend since first year
- Relevant coursework: Data Structures & Algorithms, Backend & Frontend Development, Database Management Systems, Object-Oriented Programming, Linear Algebra, Calculus 1-2
- Beyond the curriculum: preparing for the AWS Certified Solutions Architect – Associate exam; studying applied machine learning independently

PROJECTS

Full write-ups & live demos: kemalozyon.com/projects

FastAPI Blog — Python, FastAPI, async SQLAlchemy, PostgreSQL, Alembic, JWT, AWS S3, Render

- Built and deployed a full-stack blog platform serving server-rendered pages and a JSON REST API from a single app — JWT auth, fully async PostgreSQL (SQLAlchemy 2.0)
- Diagnosed and fixed real production issues independently: moved transactional email to an HTTP API after the host blocked outbound SMTP, migrated uploads to S3 after hitting the ephemeral filesystem, and resolved async lazy-loading errors with explicit eager loading
- Enforced migration discipline with Alembic — autogenerated migrations reviewed manually before production

SustainMart — Node.js, Express.js, MySQL, REST API, MVC Architecture

- Built the backend for a two-sided marketplace (CTIS 256) with distinct market and consumer roles — shared authentication, role-based authorization enforced via middleware
- Implemented two-step email verification before account creation; organized the codebase into routes/controllers/models/middleware; handled image uploads with Multer

Personal Portfolio Site — Express, MongoDB, Mongoose, JWT, Cloudinary (React frontend)

- Designed and built the backend powering kemalozyon.com — auth flow, content data models, a JWT-protected admin dashboard, and a Multer → Cloudinary image pipeline
- Directed AI to implement the React/Tailwind frontend from original designs and specs, reviewing and iterating on the generated code; backend engineering and product direction are original work

Demir Aslan — Digital Price Board — Electron, Node.js, JavaScript, IPC

- Built a cross-platform desktop app displaying real-time gold prices in a retail store — dual-window IPC sync, packaged as standalone Windows/macOS executables

Bank Management System — Java, OOP, Java Swing

- Built a GUI banking application applying core OOP principles, with modular separation between presentation and business logic across multiple account types

SKILLS

Languages: Python, JavaScript, Java, C, C++, SQL

Backend: FastAPI, Express.js, Node.js, REST API Design, JWT Auth, MVC Architecture

Databases: PostgreSQL, MySQL, MongoDB, SQLAlchemy (async), Mongoose, Alembic

Cloud & DevOps: AWS (S3; Solutions Architect Associate — in progress), Render, Docker, Git/GitHub, Linux

AI Frameworks: LangGraph, OpenAI API